

Harshit Kumar

Software Developer | Backend Development | AI/ML

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GitHub | LinkedIn

SUMMARY

Computer Science undergraduate focused on backend engineering, AI/ML, and full-stack development. Experience building REST APIs, authentication and RBAC, WebSocket systems, local LLM applications, RAG pipelines, document processing/OCR, and computer vision solutions with Python, FastAPI, React, PyTorch, and SQL/NoSQL technologies.

EDUCATION

B.Tech, Computer Science Engineering	2024–2028
GL Bajaj Institute of Technology and Management	
CBSE (Class XII)	2023
Green Valley English School	

TECHNICAL SKILLS

Languages: Python, C++, SQL
Backend: FastAPI, REST APIs, WebSockets, JWT, RBAC, SQLAlchemy
Frontend: React, JavaScript, HTML5, CSS3, Tailwind CSS, Vite
AI/ML: Machine Learning, Scikit-learn, PyTorch, YOLO, Computer Vision, OpenCV
Generative AI: Ollama, RAG, ChromaDB
Data & Document Processing: PostgreSQL, SQLite, Pandas, NumPy, PyMuPDF, Tesseract OCR
Tools: Git, GitHub, Maven, Uvicorn

PROJECTS

CollegeBuddy Full-Stack Event Management & Attendance Platform	GitHub
- Built a full-stack campus event platform with React, FastAPI, PostgreSQL/SQLite, and SQLAlchemy for student, host, and volunteer workflows. - Implemented JWT authentication and RBAC, event registration, QR ticket generation, and browser-based attendance verification. - Integrated a Scikit-learn Linear Regression model to predict event attendance from participant limits, fees, and registration data. - Added event analytics, file uploads, and REST APIs with a decoupled React frontend and FastAPI backend.	
Sovereign AI Workbench Local AI Document & Knowledge Assistant	GitHub
- Developed an offline-first AI workbench using FastAPI and Ollama for local LLM chat, streaming responses, PDF analysis, and question answering. - Built a RAG pipeline with Ollama embeddings and ChromaDB for private, session-scoped retrieval and source-aware responses. - Implemented Tesseract OCR fallback for scanned PDFs using PyMuPDF and Pillow, supporting mixed native-text and OCR workflows. - Designed bounded agent tasks for document search, calculations, and verified TXT/PDF report generation with authentication, RBAC, audit controls, and request limits.	
VIGILNET Institute Inspection & Real-Time Monitoring Platform	GitHub
- Developed a FastAPI/WebSocket telemetry system with Python agents to collect and stream CPU, memory, network, disk, and system metrics from monitored PCs. - Built React monitoring interfaces plus a Spring Boot/Java backend for authentication, institute management, inspection workflows, and ML result storage. - Integrated YOLO11s-Seg computer vision inference for structural defect detection and segmentation across crack, spalling, corrosion, and efflorescence classes. - Implemented JWT authentication, RBAC, audit logging, and PostgreSQL JSONB persistence.	

ACHIEVEMENTS

- Finalist in 4 out of 10+ hackathons.
- NPTEL Certified – Database Management Systems.
- AWS Academy Certified – Cloud Foundations.

EXTRACURRICULAR ACTIVITIES

- Member, Navrang Cultural Club, GL Bajaj Institute of Technology and Management.
- Participated in debates and group discussions (GD).